

Classic Riddler

15 April 2022

Riddle:

For this riddle, we'll define N as being a Carmichael number if (1) it is composite and has no square factors, and (2) if one less than every prime factor of N is a factor of $N - 1$.

That's a lot to take in at once, so let's look at an example. The smallest Carmichael number is 561. Sure enough, it has no square factors (other than 1, which we're not counting here). Meanwhile, its prime factors are 3, 11 and 17. If we subtract one from each of those, we get 2, 10 and 16, all of which divide 560 (one less than the original number).

Daniel's puzzle asks you to find a six-digit Carmichael number that can be written as ABCABC in base 10. (**Note:** The digits A, B and C are *not* necessarily distinct.) Of course, you could look this up in a matter of seconds or even write some code to figure it out. But I assure you, this riddle can be done by hand, so please give it a shot!

Solution:

The number ABCABC can easily be factored into $1001 \times ABC$. 1001 itself can be further factored into $7 \times 11 \times 13$. So at least 7, 11, and 13 are factors, and none of 7, 11, or 13 can divide ABC. In order to be a Carmichael number, 6, 10, and 12 must divide $ABCABC - 1$. Because $ABCABC - 1$ is divisible by 10, $C - 1 = 0$, so C must be 1. Because AB1AB0 is divisible by 3 (because it is divisible by 6), neither AB1AB1 nor AB1 is divisible by 3.

There are 60 combinations of A and B where 3 is not a factor of AB1 (101, 121, 131, 151, etc.). This is few enough that they can be checked by hand. Luckily, the lowest combination, 101 meets the Carmichael criteria. 101 itself is prime, so the prime factorization of 101,101 is $7 \times 11 \times 13 \times 101$. 101,100 must therefore be divisible by 6, 10, 12, and 100. It trivially is divisible by 10 and 100 ($= 2 \times 2 \times 5 \times 5$), as well as 3. Because of the factor of 3 and two factors of 2, it must also be divisible by 6 and 12.

So the value of 101,101 is a solution to the riddle. I don't know if there are additional solutions.